

22. Answer: D

Potential difference across upper resistor,

$$V_1 = IR$$

$$= 4 \times 30$$

$$= 120 \text{ V}$$

Current through the lower resistor, $I = 4 + 2 = 6 \text{ A}$

Potential difference across lower resistor,

$$V_2 = IR$$

$$= 6 \times 30$$

$$= 180 \text{ V}$$

Potential difference across the entire circuit $= 120 + 180 = 300 \text{ V}$

Since the lower end of the circuit is of potential 0 V, the potential at P is 300 V.